Scalability & Future Plan

Date	23 October 2026
Team ID	0827CY231053
Project Name	Waypoint - AI Travel Itinerary Planner
Maximum Marks	1 Mark

Scalability & Future Plan

Distinguish the current code and configuration from proposed improvements. Timelines below are indicative planning windows, not committed delivery dates.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Finite catalog and estimated logistics / costs.	Plans need traveler review and local verification.	High
2	No measured concurrency capacity; SQLite is a local mode.	Production capacity and database behavior need load tests.	High
3	Per-day refinement can replace a complete returned plan.	Other days may change; preservation needs a dedicated merge/check.	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	One modular FastAPI service; no capacity report.	Measure limits; add managed API instances with shared Supabase storage as needed.
2	Data Storage	Supabase when configured, SQLite locally.	Review indexes, query pagination, backup and migration procedures for production.
3	Performance	Synchronous request workflow; optional external model calls.	Measure latency; consider background generation, caching and provider retry policy.
4	Security	bcrypt, JWT and user-scoped repository calls.	Add rate limiting, improve secret rotation and review production error exposure.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Improve unchanged-day refinement; benchmark performance; broaden destination coverage.	Indicative: 1-2 months	More predictable edits and measured release readiness.
Phase 3	Multi-city planning and collaborative trip editing.	Indicative: 3-4 months	Support more complex trips and shared planning.
Phase 4	Offline itinerary access and native mobile application.	Indicative: 5-6 months	Make saved plans easier to use while traveling.

Validation Before Scale

- Run isolated read/write and generation load tests in each database mode.
- Record routine API and model latency separately, including error rates.
- Confirm public share access boundaries and review saved-day behavior after refinement.

Project sources: 1. Brainstorming and Ideation/README.md; planning/testing files; backend services, store and security modules.